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Viral Dynamics and Immune Correlates of Coronavirus Disease 2019 (COVID-19) Severity

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(See the Editorial Commentary by Sachithanandham et al on pages e2943–5.)

Background. Key knowledge gaps remain in the understanding of viral dynamics and immune response of Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) infection.

Methods. We evaluated these characteristics and established their association with clinical severity in a prospective observational cohort study of 100 patients with PCR-confirmed SARS-CoV-2 infection (mean age, 46 years; 56% male; 38% with comorbidities). Respiratory samples (n = 74) were collected for viral culture, serum samples for measurement of IgM/IgG levels (n = 30), and plasma samples for levels of inflammatory cytokines and chemokines (n = 81). Disease severity was correlated with results from viral culture, serologic testing, and immune markers.

Results. Fifty-seven (57%) patients developed viral pneumonia, of whom 20 (20%) required supplemental oxygen, including 12 (12%) with invasive mechanical ventilation. Viral culture from respiratory samples was positive for 19 of 74 patients (26%). No virus was isolated when the PCR cycle threshold (Ct) value was >30 or >14 days after symptom onset. Seroconversion occurred at a median (IQR) of 12.5 (9–18) days for IgM and 15.0 (12–20) days for IgG; 54/62 patients (87.1%) sampled at day 14 or later seroconverted. Severe infections were associated with earlier seroconversion and higher peak IgM and IgG levels. Levels of IP-10, HGF, IL-6, MCP-1, MIP-1α, IL-12p70, IL-18, VEGF-A, PDGF-BB, and IL-1RA significantly correlated with disease severity.

Conclusions. We found virus viability was associated with lower PCR Ct value in early illness. A stronger antibody response was associated with disease severity. The overactive proinflammatory immune signatures offer targets for host-directed immunotherapy, which should be evaluated in randomized controlled trials.

Keywords. COVID-19; cytokines; serology; viral culture; immunology.

The coronavirus disease 2019 (COVID-19) pandemic caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) has caused great harm to health and the global economy [1, 2]. Understanding of SARS-CoV-2 pathogeneisis has advanced at an unprecedented speed, but key gaps remain and preliminary findings require validation.

Studies of patients with COVID-19 have described the inflammatory milieu in severe infections, with increased neutrophils, suppressed lymphocytes, and elevated inflammatory mediators [3, 4]. However, most studies are limited to comparing severe against nonsevere infections and lack serial data [5, 6]. A clearer definition of disease pathogenesis will support the development of risk-stratification tools and therapeutics targeting critical pathways in the inflammatory cascade.

Early seroconversion has been reported with immunoglobulin (Ig) G to the receptor binding domain (RBD) detected at day 7 [7–10]. However, evidence of correlation between antibody titers and disease severity is conflicting [8, 10–12]. There remains a need for more detailed assessment of antibody kinetics to help determine the impact of antibody-dependent enhancement in COVID-19 pathogenesis, as well as guide convalescent plasma harvesting and use of serological assays for diagnosis [13].

SARS-CoV-2 can be detected from the nasopharynx for a median of 2–3 weeks following onset of symptoms [14]. Several studies have reported that, in immunocompetent individuals, virus is typically only cultured from respiratory samples during the first week of illness when viral loads are highest [8, 15, 16].

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[a]The full list of members of the Singapore 2019 Novel Coronavirus Outbreak Research Team is shown in Supplementary Table 4 in the Supplementary Appendix.

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This suggests transmission risk declines in the second week. This finding requires confirmation in larger cohorts as it has important implications for infection-control and isolation protocols [17].

In this multipronged study, we describe the serologic evolution, inflammatory response, and pattern of viral shedding and viability in patients with virologically confirmed COVID-19 in Singapore, and analyze the contributions these make to severe infections.

METHODS

Patient Recruitment

All individuals confirmed to have COVID-19 by SARS-CoV-2 real-time reverse transcriptase–polymerase chain reaction (RT-PCR) and admitted to any of 7 public hospitals in Singapore were eligible for inclusion in this study. RT-PCR was performed on respiratory samples, as previously described [18] (details shown in the Supplementary Methods).

Data and Specimen Collection

Clinical information was extracted from the medical record using a standardized data-collection form adapted from the International Severe Acute Respiratory and Emerging Infection Consortium (ISARIC) case record form [19]. Serial blood and respiratory samples were collected during hospitalization and follow-up postdischarge (days 1, 3, 7, 14, 21, and 28 after enrollment). A stool and urine sample was also collected for SARS-CoV-2 PCR and culture on day 1.

Clinical Management

At the time of the study, all patients with COVID-19 in Singapore were admitted to airborne infection–isolation rooms regardless of disease severity. Supportive therapy, including supplemental oxygen and symptomatic treatment, was administered as required. Patients with moderate to severe hypoxia were transferred to the intensive care unit (ICU) for further management and invasive mechanical ventilation as required.

Patients were discharged from the hospital only after resolution of symptoms and when 2 consecutive nasopharyngeal swabs more than 24 hours apart were negative for SARS-CoV-2 by RT-PCR. Follow-up visits were arranged on day 28 of enrollment.

Virus Isolation

Material from nasopharyngeal swabs was collected in universal transport media and used to inoculate Vero-E6 cells (American Type Culture Collection: CRL-1586TM) for virus isolation in an Animal Biological Safety Level 3 laboratory. Urine and stool samples were collected and transported fresh for viral culture. Stools were filtered before inoculation. Cells were cultured at 37°C for 7 days or less if cytopathic effect (CPE) was observed, and 3 blind passages were performed. Cytopathic effect consisted of rounded cells and extensive cell death, usually by day 4 postinoculation. Positive isolation was confirmed by the observation of CPE and virus-specific PCR. Total RNA was extracted from all samples using the E.Z.N.A. Total RNA Kit I (Omega Bio-Tek, Inc) according to the manufacturer's instructions, and samples were analyzed by RT-PCR for the detection of SARS-CoV-2 as previously described [20].

Capture Enzyme-linked Immunosorbent Assay to Detect IgM and IgG

Serum collected during the acute and convalescent phases of infection were tested for SARS-CoV-2 receptor binding domain–specific IgM and IgG using capture enzyme-linked immunosorbent assay (ELISA) (details in the Supplementary Methods).

Multiplex Microbead-based Immunoassay

Plasma samples were collected during acute and convalescent phases and treated with a solvent/detergent based on Triton X-100 (1%) (Dow Chemical Company) for virus inactivation [21]. Immune mediator levels were measured using Cytokine/Chemokine/Growth Factor 45-Plex Human ProcartaPlex Panel 1 (ThermoFisher Scientific) (details in the Supplementary Methods). Cytokine levels were also measured in 23 healthy donor plasma samples as baseline controls.

Statistical Analysis

Patients were categorized into 3 groups for comparison: no pneumonia on chest radiographs (CXR) throughout admission, pneumonia on CXR without hypoxia, and pneumonia with hypoxia (desaturation to ≤94%) needing supplemental oxygen. Day 1 was defined as the first day of symptom onset.

Fold-changes (FCs) of antibody titers compared with negative controls were calculated. Kruskal-Wallis test followed by Dunn's multiple-comparison test was used to determine significance of antibody levels.

Unpaired *t* test was applied to ascertain significant differences in the immune mediator levels between the patients with COVID-19 and healthy controls at different time points post–illness onset. One-way analysis of variance (ANOVA) with post hoc *t* test corrected using the method of Bonferroni was used to discern the differences in immune mediator levels between the various disease-severity groups. One-way ANOVA results were corrected for multiple testing using the method of Benjamini and Hochberg.

TM4-MeV Suite (version 10.2) was used to compute hierarchical clustering and heat map on the immune mediators (details in Supplementary Methods). Biological processes and immune pathways were predicted and illustrated using the Search Tool for the Retrieval of Interacting Genes/Proteins database (STRING; version 11.0; available at: https://string-db.org). All interactions were derived from high-throughput laboratory experiments and previous knowledge in curated databases at a confidence threshold of 0.8.

Student's t test or Mann-Whitney U test was used as appropriate for continuous variables and Fisher's exact test for categorical variables. P values less than .05 were considered statistically significant. Plots were generated using GraphPad Prism (version 7), MedCalc Statistical Software (version 19.2.1), or R (version 3.3.1).

Ethical Approval

Written informed consent was obtained from all participants. The study protocol at all sites was approved by the National Healthcare Group Domain Specific Review Board, study reference 2012/00917; additional study protocol at Singapore General Hospital was approved by the SingHealth Centralized Institutional Review Board, study reference 2018/3045.

RESULTS

Clinical Presentation, Treatment, and Outcome

A total of 130 patients with COVID-19 were diagnosed in Singapore from the first known case on 22 January 2020 to 6 March 2020 (Figure 1). Among the 100 (77%) enrolled in this study, viral pneumonia was diagnosed by CXR in 57, of whom 20 required supplemental oxygen for hypoxia and 12 required invasive mechanical ventilation (Table 1).

Following resolution of infection and viral shedding, 97 patients have been discharged, and after 90 days of follow-up no reinfection or recrudescence has been detected. Three patients have died. Deaths occurred 27 days or more after hospital admission, and while all were assessed by the managing clinician as related to COVID-19, they followed viral clearance and prolonged invasive mechanical ventilation.

Viral Shedding and Cultures

SARS-CoV-2 was detectable from nasopharyngeal swabs by PCR up to 48 days after symptom onset. The mean duration of viral shedding by PCR was 16.7 days (95% confidence interval [CI], 15.2–18.3 days). Cessation of viral shedding by PCR occurred in 4% by day 7, 30% by day 14, 78% by day 21, and 91% by day 28. There were no differences in the duration of viral shedding stratified by disease severity (Supplementary Figures 1 and 2).

Viral culture was attempted from 152 respiratory samples where SARS-CoV-2 was detected by PCR. These samples were collected from 74 patients and SARS-CoV-2 was isolated from 21 samples from 19 (26%) patients (Supplementary Table 1). No virus was isolated from samples where the cycle threshold (Ct) value was greater than 30, or when the sample was collected more than 14 days after symptoms onset (Figure 2, Supplementary Figures 2 and 3). There was no correlation between virus isolation by culture and infection severity, patient demographics, or symptomatology (data not shown). Viral culture was attempted from the stool of 34 patients and urine of 24 patients during acute infection; in all patients, virus was not isolated. Seven (21%) patients had virus detectable by PCR in stool.

IgM and IgG Titers Correlate With Disease Severity

Serial serum samples for ELISA were available for 30 patients. Anti–SARS-CoV-2 IgM was first detectable in 17.9% in the first week of illness, 39.3% in the second week, 35.7% in the third week, and 7.1% after the third week, and correspondingly for IgG, 7.7%, 26.9%, 50.0%, and 15.4%. The median time to seroconversion was 12.5 days (interquartile range [IQR], 9–18 days)

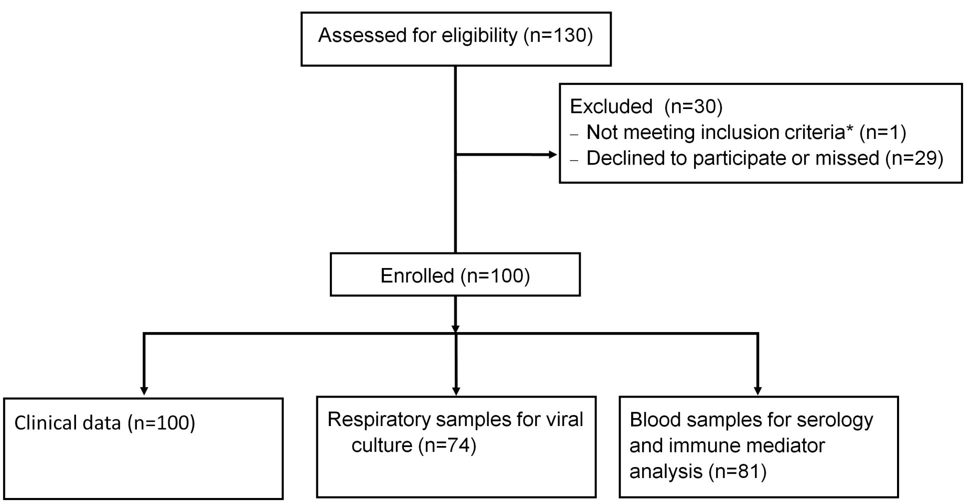

Figure 1. Flow diagram of study recruitment and follow-up in 130 patients diagnosed in Singapore as of 6 March 2020. *One excluded as diagnosis was retrospective via serology rather than PCR. The 100 patients were enrolled from the following hospitals in Singapore: National Centre for Infectious Diseases (77), National University Hospital (8), Singapore General Hospital (8), Ng Teng Fong General Hospital (3), Changi General Hospital (2), Alexandra Hospital (1), Khoo Teck Puat Hospital (1). Abbreviation: PCR, polymerase chain reaction.

Table 1. **Demographics, Presenting Symptoms, Parameters, and Laboratory Investigations Stratified by Disease Severity**

	All (N = 100)	No Pneumonia (Group A) (n = 43)	Pneumonia but No Hypoxia (Group B) (n = 37)	*P* (A vs B)	Pneumonia and Hypoxia (Group C) (n = 20)	*P* (A vs C)
Demographics						
Age, years	44 (35–56)	37 (31–50)	44 (35–55)	.080	62 (49–68)	<.0001
Sex, male, n (%)	56 (56)	23 (53)	19 (51)	1.0000	6 (70)	.28
Ethnicity (Chinese), n (%)	83 (83)	33 (77)	31 (84)	.58	19 (95)	.15
Comorbidity, any, n (%)	38 (38)	12 (28)	14 (38)	.47	12 (60)	.025
Diabetes, n (%)	10 (10)	2 (5)	3 (8)	.66	5 (15)	.033
Hypertension, n (%)	19 (19)	2 (5)	8 (22)	.039	9 (45)	.0003
Charlson's score	0 (0–0)	0 (0–0)	0 (0–0)	.25	0 (0–1)	.0067
Presenting symptoms						
Duration of symptoms before admission, days	6 (2–9)	5 (2–9)	6 (4–9)	.24	7 (2.4–8)	.65
Fever, n (%)	76 (76)	26 (60)	31 (84)	.027	19 (95)	.0059
Cough, n (%)	70 (70)	29 (67)	25 (68)	1.000	16 (80)	.38
Dyspnea, n (%)	17 (17)	3 (7)	6 (16)	.76	8 (40)	.022
Sore throat and/or rhinorrhea, n (%)	47 (47)	29 (67)	17 (46)	.070	6 (30)	.0071
Diarrhea, n (%)	19 (19)	9 (21)	6 (16)	.78	4 (20)	1.000
Baseline parameters						
Temperature, °C	37.6 (37.0–38.3)	37.4 (36.9–38.0)	37.8 (37.0–38.2)	.26	38.5 (37.7–39.0)	.0045
Respiratory rate, breaths/minute	18 (17–19)	18 (17–18)	18 (17–18)	.31	20 (19–20)	.0002
Heart rate, beats/minute	89 (77–99)	85 (75–98)	86 (76–105)	.53	95 (86–104)	.017
Systolic blood pressure, mm Hg	132 (120–144)	131 (121–139)	132 (121–146)	.76	131 (112–144)	.74
Pulse oximeter oxygen saturation, %	98 (96–99)	98 (98–100)	98 (97–99)	.20	96 (95–97)	<.0001
Baseline laboratory investigations						
Hemoglobin, g/dL	13.9 (12.9–15.1)	13.9 (13.1–15.2)	14.4 (13.3–15.3)	.33	13.1 (12.3–14.2)	.033
Platelets, x 10^9/L	193 (151–259)	213 (183–263)	185 (144–259)	.14	157 (134–198)	.0017
Lymphocytes, x 10^9/L	1.1 (0.8–1.5)	1.4 (0.9–1.9)	1.1 (0.8–1.5)	.070	0.5 (0.8–1.1)	.0001
Neutrophils, x 10^9/L	3.0 (2.3–4.1)	2.6 (2.3–4.1)	3.1 (2.2–3.7)	.97	3.2 (2.4–5.1)	.32
CRP (n = 81), mg/L	11.4 (3.3–47.4)	2.8 (1.1–9.4)	11.9 (7.1–22.1)	<.0001	87.9 (64.6–153.1)	<.0001
LDH (n = 90), U/L	443 (365–595)	368 (342–416)	463 (389–585)	.0024	632 (525–871)	<.0001
Creatinine (n = 97), µmol/L	67 (54–83)	65 (53–79)	69 (52–81)	.78	76 (65–92)	.038
ALT (n = 85), U/L	34 (28–40)	23 (15–32)	27 (15–44)	.27	36 (30–56)	.0014
PCR Ct value at diagnosis	28.2 (24.3–33.3)	28.0 (21.3–34.5)	29.3 (25.4–33.9)	.35	27.0 (24.8–31.2)	.94

Values are medians (IQR) or n (%). Groups were compared using Mann-Whitney *U* test or Fisher's exact test.
Abbreviations: ALT, alanine aminotransferase; CRP, C-reactive protein; Ct, cycle threshold; IQR, interquartile range; LDH, lactate dehydrogenase; PCR, polymerase chain reaction.

for IgM and 15.0 days (IQR, 12–20 days) for IgG. The total seroconversion rate from 62 patients sampled on day 14 or later was 87.1%. Time to IgG seroconversion was significantly shorter for patients with pneumonia and hypoxia compared with less-severe infections (median, 12.5 vs 18 days; *P* = .013, Mann-Whitney). There was also a significant correlation between disease severity and peak IgM and IgG levels (Figure 3). However, IgM and IgG antibody levels were not associated with duration of viral shedding (Figure 3), baseline viral Ct values from respiratory samples, age, or comorbidities (Supplementary Figure 5).

Proinflammatory Signature of Severe COVID-19

A total of 321 plasma samples were available from 81 patients. Levels of interleukin (IL)-2, IL-18, interferon-γ (IFN-γ), tumor necrosis factor α (TNF-α), monocyte chemoattractant protein 1 (MCP-1), hepatocyte growth factor (HGF), brain-derived neurotrophic factor (BDNF), leukemia inhibitory factor (LIF), placenta growth factor-1 (PLGF-1), and beta-nerve growth factor (bNGF) were significantly elevated compared with healthy controls for their first plasma samples taken soon after admission (Figure 4A). Proinflammatory cytokines and chemokines (IL-2, IL-18, IFN-γ, and MCP-1) were elevated at the early phase of infection and were reduced to a healthy basal range at approximately 15 days post–illness onset. In contrast, the elevated levels of lung injury–associated growth factors (HGF, BDNF, LIF, PLGF-1, and bNGF) persisted throughout the infection course (Figure 4A). Stratification of patients with COVID-19 with plasma samples nearest to the onset of pneumonia and hypoxia revealed that the proinflammatory cytokines IFN-γ inducible protein (IP-10), HGF, IL-6, MCP-1, macrophage inflammatory protein-1α (MIP-1α), IL-12p70, and IL-18; growth

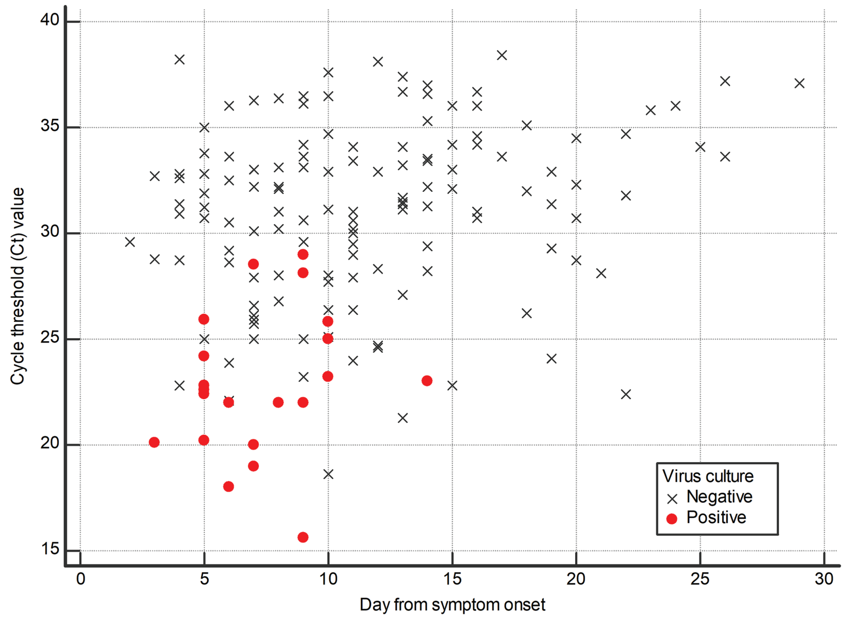

Figure 2. Scatterplot of viral culture results by day from symptom onset and PCR Ct value. Analysis includes 152 samples from 74 patients in whom virus was detectable by SARS-CoV-2 PCR. Virus was cultured from 21 (14%) samples from 19 patients. Abbreviations: PCR, polymerase chain reaction; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2.

factors vascular endothelial growth factor A (VEGF-A) and 2 B subunits of platelet-derived growth factor (PDGF-BB); and anti-inflammatory cytokine IL-1RA were associated with severity. Levels were significantly elevated between patients with pneumonia without hypoxia and patients with hypoxia compared with patients without pneumonia, except for MCP-1, which was only significantly higher in patients with hypoxia (Figure 4B and 4C; Supplementary Tables 2 and 3).

Further network analyses showed an association between the aforementioned immune mediators and clinical parameters (pneumonia, hypoxia, and ICU admission). The intertwined relationship of the cytokines is shown in Figure 5A. In addition, Ingenuity Pathway Analysis (IPA) revealed canonical pathways associated with these immune mediators and severity, with the top 10 canonical pathways involved in inflammatory diseases and cell signalling (Figure 5B). The top canonical pathway highlights the common immune mediators between influenza and SARS-CoV-2 infection, including MCP-1, IL-1RA, IP-10, and IL-6 (Figure 5C). In addition to IPA, STRING prediction of protein–protein interactions identified IL-6 as a direct interacting partner with other severity-associated immune mediators (Figure 5D).

A longitudinal comparison of these immune mediators associated with severity in 12 ICU patients was performed to explore their role as prognostic markers for severe COVID-19 (Supplementary Figure 6). Four of these 12 patients (CT009, CT032, CT037, and CT057) recovered and were discharged at the time of study. Interestingly, HGF and VEGF-A were distinctly separated into 2 levels, with the 4 discharged patients having lower HGF and VEGF-A levels, and HGF approaching healthy baseline levels during convalescence, which further indicates that high levels of these cytokines were associated with poor prognosis in ICU patients. Similarly, these patients had lower levels of MCP-1 and IL-6 compared with other ICU patients despite having a less distinct separation. Longitudinal comparison of other immune mediators, including IP-10, IL-18, PDGF-BB, and IL-1RA, revealed decreasing levels with days post–ICU admission and approaching healthy baseline levels during the latter period after ICU admission (Supplementary Figure 6).

DISCUSSION

This study of 100 patients in the first few months of the COVID-19 pandemic in Singapore provides a detailed overview of clinical presentation, progress, and outcomes. Case detection and contact tracing in Singapore are rigorous, with the main gap possibly in missing asymptomatic cases [22]. From the cohort, 43% of patients never developed pneumonia, 37% developed pneumonia without hypoxia, and 20% developed pneumonia with hypoxia. We found no relationship between illness severity and duration of viral shedding or PCR Ct values. The central role of the immune response to SARS-CoV-2 in COVID-19 was evident from the strong correlation between disease severity and levels of IgG/IgM and inflammatory immune mediators in our cohort.

Viral shedding of SARS-CoV-2 from the respiratory tract has been observed to persist for several weeks and potentially up

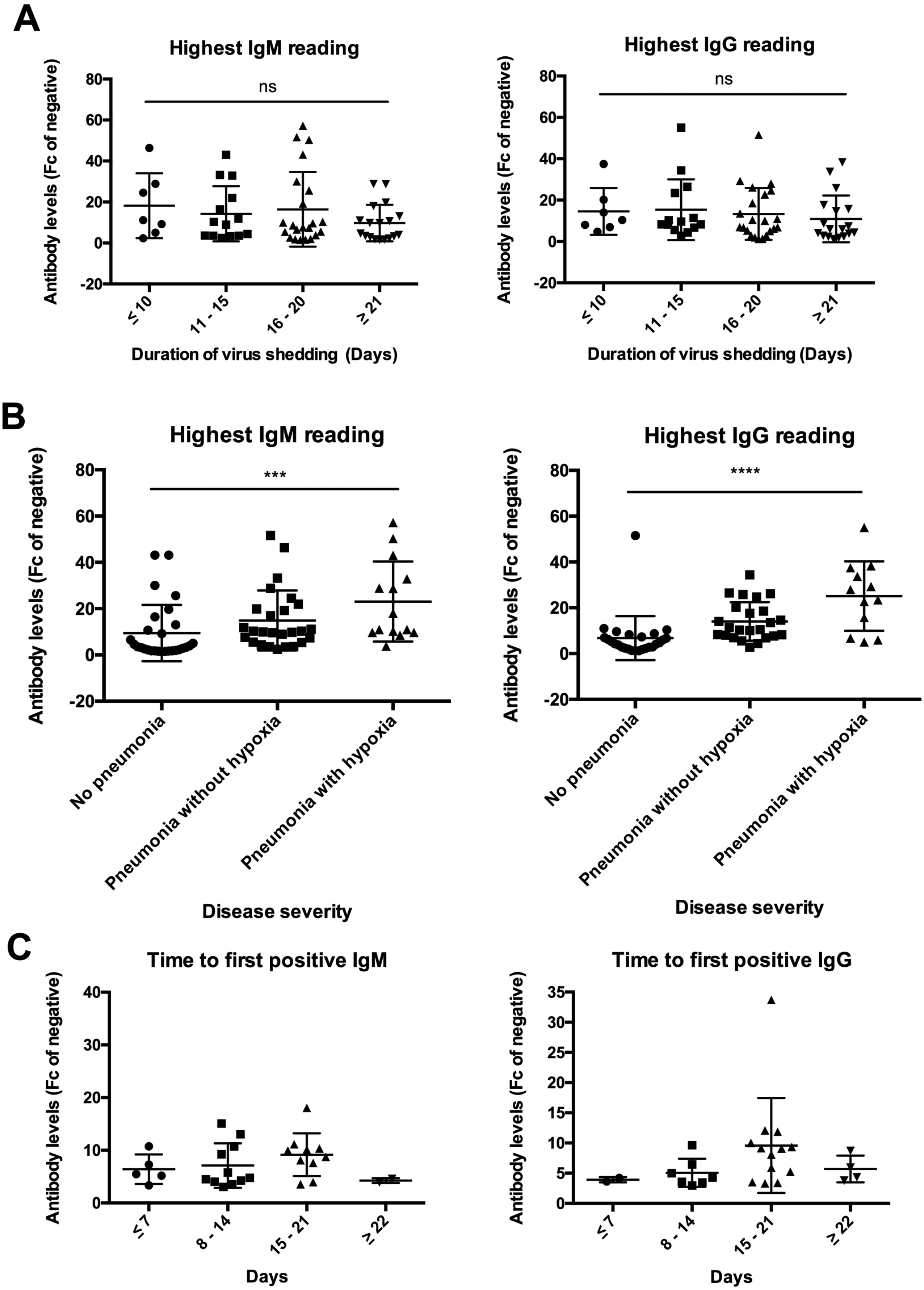

Figure 3. IgG and IgM readings stratified by (*A*) duration of viral shedding, (*B*) disease severity, and (*C*) time to first positive antibody level. Fc is calculated by dividing OD reading of a test sample by the average OD reading of negative controls. Samples with Fc >3 are considered positive. As sampling time and numbers were not uniform for all patients, we plotted the highest IgM/IgG from a single serum sample for each patient when multiple samples were available. Abbreviations: Fc, fold-change; IgG, immunoglobulin G; IgM, immunoglobulin M; ns, not significant; OD, optical density.

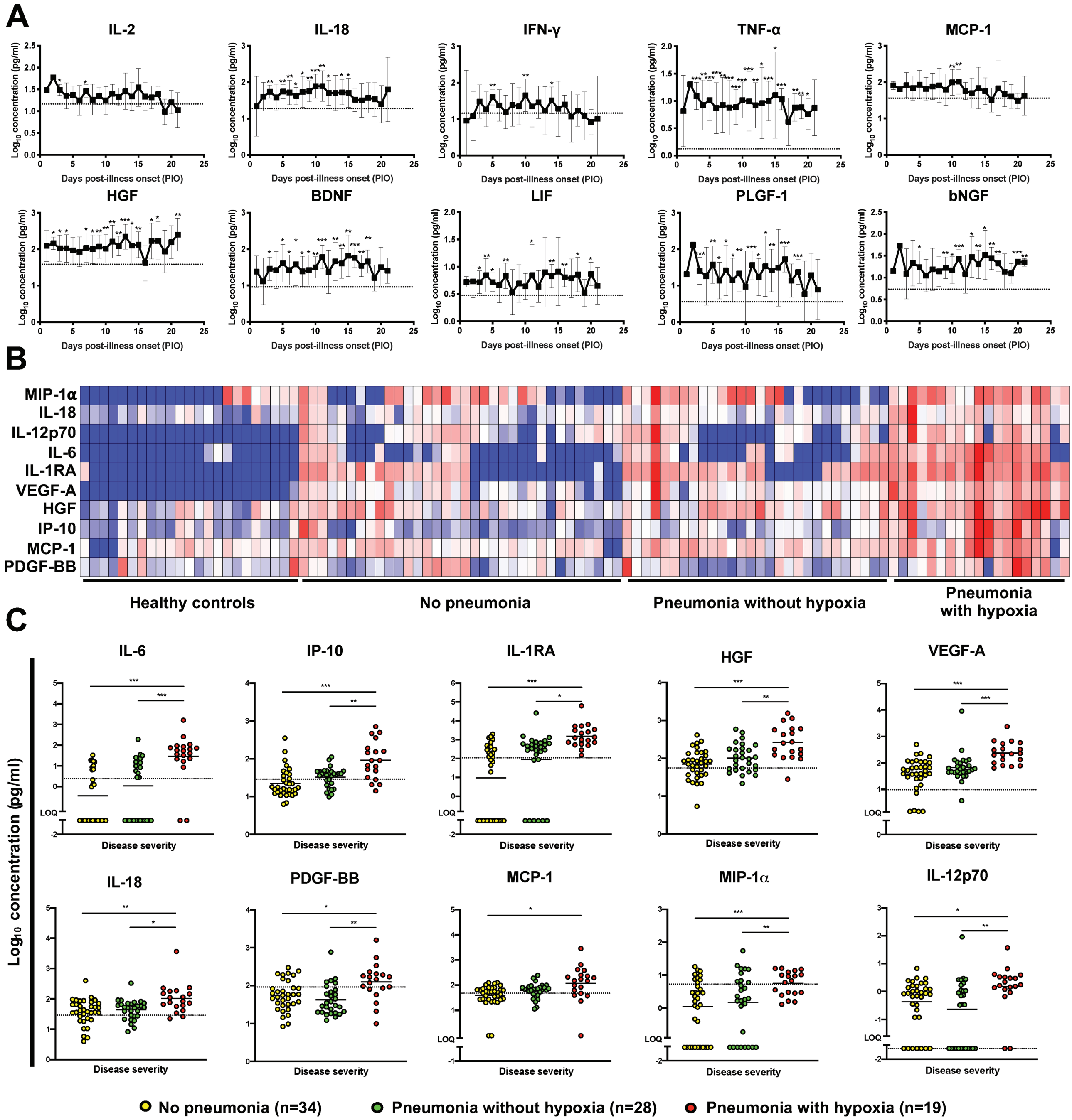

Figure 4. Immune signatures of patients with COVID-19 reveal cytokines associated with disease severity. Plasma fractions were isolated from the blood of patients with COVID-19 (n = 81) at different time points. Time points closest to event onset, including mild (no pneumonia), pneumonia without hypoxia, or pneumonia with hypoxia, were chosen to identify cytokines that were associated with disease severity. Concentrations of 45 immune mediators were quantified using a 45-plex microbead-based immunoassay. *A*, Longitudinal profile of detectable cytokines in patients with COVID-19 during the acute phase of disease. Statistical analyses were performed using unpaired *t* test against the healthy baseline (* = $P < .05$, ** = $P < .01$, *** = $P < .001$). Cytokine level for healthy controls (n = 23) is indicated by the dotted line. *B*, Heatmap of severity-associated cytokine levels in patients with different disease outcomes (healthy controls, n = 23; no pneumonia, n = 34; pneumonia without hypoxia, n = 28; pneumonia with hypoxia, n = 19). Each color represents the relative concentration of a particular analyte. Blue and red indicate low and high concentrations, respectively. *C*, Profiles of significant immune mediators are illustrated as scatterplots. One-way ANOVAs were conducted on the logarithmically transformed concentration with post hoc *t* tests corrected using the method of Bonferroni. ANOVA results were corrected for multiple testing using the method of Benjamini and Hochberg (* = $P < .05$, ** = $P < .01$, *** = $P < .001$). Cytokine level for healthy controls (n = 23) is indicated by the dotted line. Patient samples that are not detectable are assigned the value of logarithm transformation of LOQ. Abbreviations: ANOVA, analysis of variance; BDNF, brain-derived neurotrophic factor; bNGF, beta-nerve growth factor; COVID-19, coronavirus disease 2019; HGF, hepatocyte growth factor; IFN-γ, interferon-γ; IL, interleukin; IL-1RA, interleukin-1 receptor antagonist; IP-10, IFN-γ inducible protein; LIF, leukemia inhibitory factor; LOQ, limit of quantification; MCP-1, monocyte chemoattractant protein-1; PDGF-BB, 2 B subunits of platelet-derived growth factor; PLGF-1, placenta growth factor-1; TNF-α, tumor necrosis factor α; VEGF-A, vascular endothelial growth factor A.

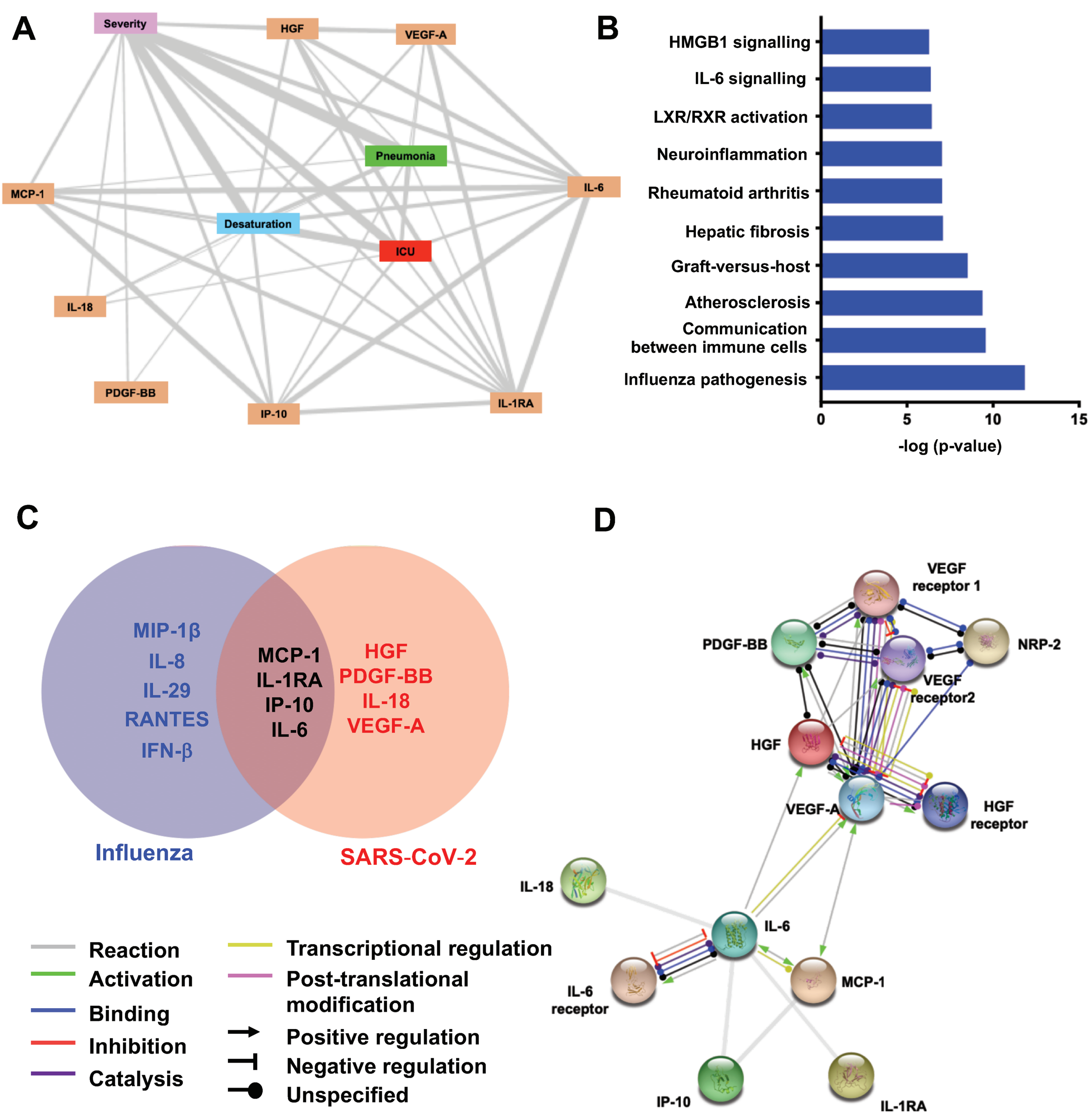

Figure 5. Network analysis of immune mediators associated with disease severity in patients with COVID-19. *A*, Interactomic analysis by IPA software. The network is displayed graphically as nodes and edges. Each node represents either severity-associated cytokine or disease condition. Thickness of edges represents the magnitude of biological relationship between connected nodes. Edges are thicker when the association between parameters is stronger. *B*, IPA analysis of the 8 significant immune mediators associated with disease severity in patients with COVID-19. The chart represents the top 10 significantly associated canonical pathways with the immune mediators. *C*, Venn diagrams show common immune mediators between influenza virus infection and SARS-CoV-2 virus infection. *D*, Interactive relationships between the immune mediators were determined by STRING analysis, with a confidence threshold of 0.8. Abbreviations: COVID-19, coronavirus disease 2019; HGF, hepatocyte growth factor; ICU, intensive care unit; IFN, interferon; IL, interleukin; IL-1RA, interleukin-1 receptor antagonist; IPA, Ingenuity Pathway Analysis; IP-10, IFN-γ inducible protein; MCP-1, monocyte chemoattractant protein-1; NRP-2, neuropilin 2; PDGF-BB, 2 B subunits of platelet-derived growth factor; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2; STRING, Search Tool for the Retrieval of Interacting Genes/Proteins database; VEGF-A, vascular endothelial growth factor A.

to months [18, 23–25]. Whether the virus remains infectious throughout this prolonged viral shedding is an important question to resolve. Smaller studies have found that viable virus was readily isolated from immunocompetent individuals during the first week of illness but were unable to successfully isolate viruses in culture from day 8 onward despite detectable viral loads by PCR [8, 25, 26]. However, a case report from Taiwan showed that it was possible to culture the virus up until day 18 [27], while positive cultures up to day 20 have been reported from a study of patients with severe infection [15]. We found that successful viral culture was associated with PCR Ct values of 30 or less. Virus was isolated up to day 14 post–symptom onset, although the majority were cultured at day 10 or earlier. Using the PCR Ct value to guide decision making on de-isolation may be an alternative to using day of illness and to provide an additional level of reassurance. However, this requires further validation.

We observed seroconversion of IgM at a median of 12.5 days and IgG at 15.0 days. Similar to a Hong Kong study of 16 patients, there were no significant differences in IgM and IgG titers by age or comorbidities [10]. We found IgM and IgG titers to be correlated with disease severity, which was similarly found in 2 Chinese studies of 173 [7] and 285 [28] patients, respectively.

We found a lower seroconversion rate of 87.1% in 62 patients from day 14 or later. A Chinese study of 173 patients reported that 94.3% of patients were IgM positive by days 15–39 and 79.8% were IgG positive by days 15–39 [7]. A Hong Kong study reported IgM positivity of 94% after day 14 and IgG positivity of 100% after day 14 in a smaller number of 16 patients [10]. The largest study from China of 285 patients reported IgM positivity of 94% by days 20–22 and IgG positivity of 100% by days 17–19 [28]. Different sample sizes and antibody assays may account for differing seroconversion rates, which merits further investigation in a large cohort over a longer period.

Serological testing is vital for determining the prevalence of infection in sero-surveys and as part of epidemiological investigations to understand transmission of clusters [29]. The timing of seroconversion may guide the timing of plasmapheresis for convalescent plasma [30, 31]. The role of antibodies in long-term immunity after infection needs further investigation.

Consistent with severe acute respiratory syndrome (SARS) and Middle East respiratory syndrome (MERS), SARS-CoV-2 infection triggers a cytokine storm in a subset of patients with markedly increased levels of proinflammatory cytokines, chemokines, and growth factors [32, 33]. Notably, higher levels of IL-6, MCP-1, IP-10, IL-18, IL-1RA, PDGF-BB, HGF, VEGF-A, IL-12p70, and MIP-1α were associated with severe disease in our study. This robust induction of proinflammatory mediators indicates that innate immune cell responses and antiviral T-cell responses are responsible for SARS-CoV-2 pathogenesis in patients with COVID-19 [34]. In addition, elevation in growth factors, including HGF [35], PLGF-1 [36], and LIF [37], illuminates the repair mechanisms following acute lung injury during SARS-CoV-2 infection. Previous studies showed that ICU patients had more significant cytokine activation compared with non-ICU patients [1]. Our longitudinal cytokine profiling in ICU patients further evaluated prognostic values of specific cytokines. We observed better prognosis in critically ill patients with lower levels of acute lung injury–associated growth factors HGF and VEGF-A at the time of admission to the ICU. Our data suggest that the differences in degree of lung injury could reflect recovery rate of patients in the ICU. HGF and VEGF-A may serve as early indicators of poor prognosis and may provide guidance to make pre-emptive clinical decisions in critically ill patients.

A central role for IL-6 in lung injury has been postulated, with higher levels associated with mortality in 2 separate studies and severe immune-mediated injury in lung tissues of patients who died of COVID-19 [38, 39]. Similarly, the IL-1 pathway has been highlighted to contribute towards SARS-CoV-2 pathogenesis [40]. Importantly, our study highlighted the upregulation of IL-1 pathway mediators IL-18, MCP-1, and VEGF-A in critically ill patients, providing further evidence that dysregulation in the IL-1 pathway could contribute to the hyperinflammatory state, especially in fatal cases.

STRING analysis revealed potential protein-to-protein interactions in severe COVID-19 infection, in which IL-6 is the direct interacting partner of other cytokines associated with disease severity. Thus, several approved IL-6 receptor antagonists could be repurposed to treat severe SARS-CoV-2 infection. Given the elevated level of VEGF-A, Janus kinase (JAK) inhibitors should also be investigated as a potential therapeutic option [41, 42]. However, the serial data presented here indicate that the therapeutic window for intervention is narrow. Immune modulators will need to be active before the inflammatory cascade causing acute lung injury develops.

Our study has several limitations. Active case finding through contact tracing is likely to have identified the majority of symptomatic infections in Singapore over this time period. However, atypical or subclinical infections may have been missed. No infections were detected in children (<18 years) or residents of long-term-care facilities. Individuals were enrolled as soon as possible following admission, but while biological samples were collected serially, they were not all acquired at the same time point after symptom onset. Additionally, some laboratory data were incomplete, and clinical data such as date of symptom onset are subject to recall bias. We also did not investigate the effect of different SARS-CoV-2 lineages or mutations on study outcomes [43]. Finally, samples were processed consistently, but numerous factors determine the success of viral cultures and the correlation between culturability and infectiousness is unclear.

In conclusion, we found that virus viability was associated with lower PCR Ct values in early illness. This merits further investigation in terms of infectivity and infection control. SAR-CoV-2 IgM and IgG did not appear until days 15–21 of illness, with implications for the role of rapid diagnostic antibody assay and the timing of plasmapheresis for convalescent plasma. A stronger antibody response was associated with disease severity, suggesting a role in immune pathogenesis. Finally, the overactive proinflammatory immune signatures offer targets for host-directed immunotherapy, which should be evaluated in randomized controlled trials.

Supplementary Data

Supplementary materials are available at *Clinical Infectious Diseases* online. Consisting of data provided by the authors to benefit the reader, the posted materials are not copyedited and are the sole responsibility of the authors, so questions or comments should be addressed to the corresponding author.

Notes

Author contributions. B. E. Y., Y.-S. L., and D. C. L. designed the study protocol. B. E. Y., S. W. X. O., P. Y. C., S. K., L. Y. A. C., S. P., S. W. T., L. S., P. P., Y. D., P. T., and J. G. H. L. collected the data. L. F. P. N., D. E. A., W. N. C., T.-M. M., S.-W. F., Y.-H. C., C. W. T., B. L., O. R., L. C., T. B., R. T. P. L., L. R., and L.-F. W. conducted the laboratory investigations including serologic, immunologic, and viral isolation. B. E. Y. and L. W. A. conducted the data analysis. B. E. Y., S. W. X. O., L. F. P. N., W. N. C., and D. C. L. drafted the manuscript. Y.-S. L., L. R., L.-F. W., and D. C. L. provided overall supervision. B. E. Y. had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. All authors read and approved the final manuscript. The Singapore 2019 Novel Coronavirus Outbreak Research team: Poh Lian Lim, Brenda Sze Peng Ang, Cheng Chuan Lee, Lawrence Soon U Lee, Li Min Ling, Oon Tek Ng, Monica Chan, Kalisvar Marimuthu, Shawn Vasoo, Chen Seong Wong, Tau Hong Lee, Sapna Sadarangani, Ray Junhao Lin, Mucheli Sharavan Sadasiv, Deborah Hee Ling Ng, Chiaw Yee Choy, Glorijoy Shi En Tan, Yu Kit Tan, Stephanie Sutjipto, Pei Hua Lee, Jun Yang Tay, Tsin Wen Yeo, Bo Yan Khoo, Woo Chiao Tay, Gabrielle Ng, Yun Yuan Mah, Wilnard Tan, Partha Pratim De, Rao Pooja, Jonathan WZ Chia, Yuan Yi Constance Chen, Shehara Mendis, Boon Kiat Toh, Raymond Kok Choon Fong, Helen May Lin Oh, Jaime Mei Fong Chien, Humaira Shafi, Hau Yiang Cheong, Thean Yen Tan, Thuan Tong Tan, Ban Hock Tan, Limin Wijaya, Indumathi Venkatachalam, Ying Ying Chua, Benjamin Pei Zhi Cherng, Yvonne Fu Zi Chan, Hei Man Wong, Siew Yee Thien, Kenneth Choon Meng Goh, Shireen Yan Ling Tan, Lynette Lin Ean Oon, Kian Sing Chan, Li Lin, Douglas Su Gin Chan, Say Tat Ooi, Deepak Rama Narayana, Jyoti Somani, Jolene Ee Ling Oon, Gabriel Zherong Yan, David Michael Allen, Roland Jureen, Benedict Yan, Randy Foo, Adrian Kang, Velraj Sivalingam, Wilson How, Norman Leo Fernandez, Nicholas Kim-Wah Yeo, Rhonda Sin-Ling Chee, Siti Naqiah Amrun.

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